

## Section 01



# Industrial Security (Bachelor)

Section 01 – Introduction to Industrial Security

## Introduction to Industrial Security

*Lecture Notes*

Department of Computer Science

- 1 Cyber-Physical Foundations
- 2 Context and History
- 3 Threat Setting and Misconceptions

## By the end of this section you will be able to

- Define Operational Technology (OT) and contrast it with Information Technology (IT).
- Identify the core components of an Industrial Control System (ICS).
- Explain why the CIA triad is re-ordered for industrial environments.
- Place industrial security in its historical and economic context.
- Recognise common misconceptions about ICS security.

# 1

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## Cyber-Physical Foundations

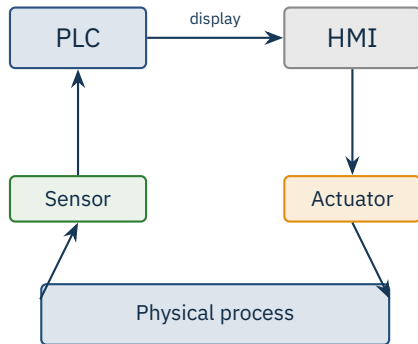
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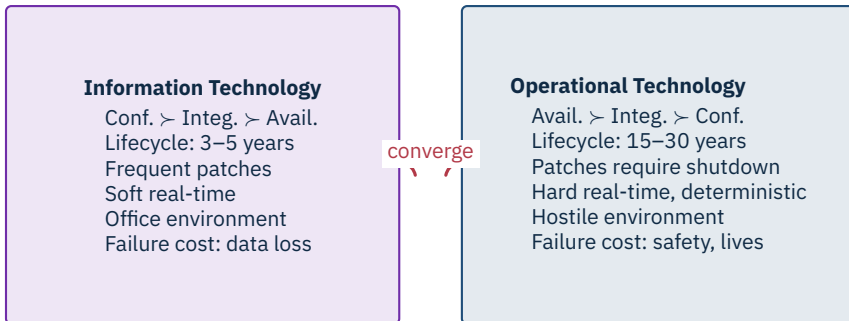
## Definition

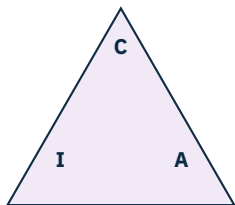
**Industrial security** is the protection of cyber-physical systems — computers that monitor and control machines in the physical world.

**Sectors:** power, water, oil & gas, manufacturing, transport, building automation.

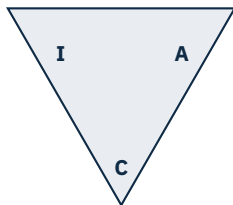
A breach has consequences in *atoms*, not just in *bits*.







IT priorities



OT priorities

## Remember

In OT, **availability** and **safety** dominate the CIA triad. A blocked alarm can blind an operator to a runaway reactor.

# 2

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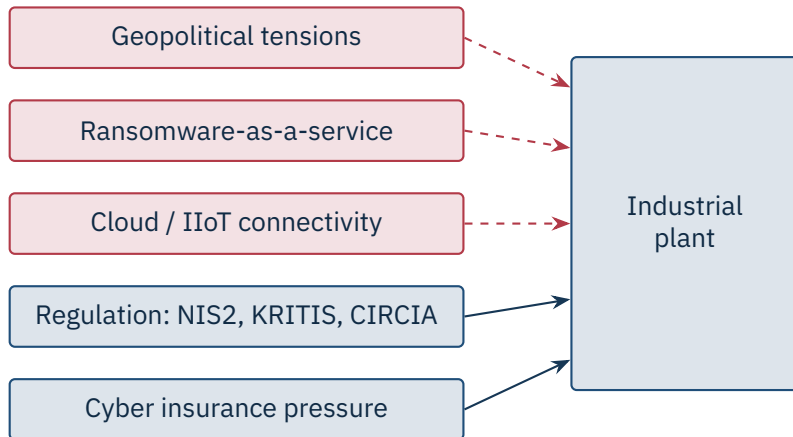
## Context and History

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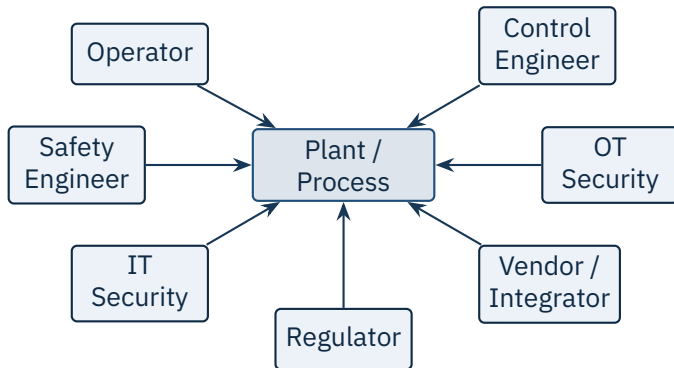




Each milestone made systems more capable – and more reachable.



Source: Dragos Year in Review 2024: 23 active threat groups tracked, ICS ransomware up 87% YoY | Munich Re Cyber Insurance Risk Snapshot, 2024.



Industrial security is a team sport across roles that rarely sat together historically.

# 3

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## Threat Setting and Misconceptions

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*“We’re air-gapped, so we’re safe.”*

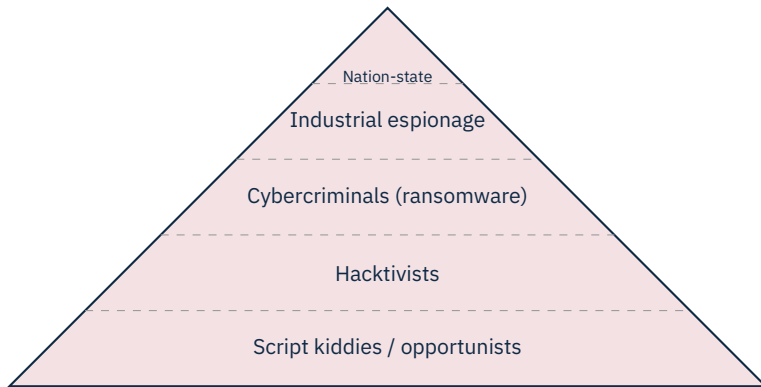
*“Our protocols are obscure; nobody understands them.”*

*“We’ve never been attacked, so we’re fine.”*

*“IT tools will solve OT problems.”*

*“Patching solves everything.”*

*“Compliance equals security.”*



Capability rises as we move up; volume of actors falls.

## Colonial Pipeline

\$4.4M ransom  
5-day outage

## Norsk Hydro

\$71M loss  
6-month recovery

## Maersk (NotPetya)

\$300M loss  
10 days down

## Stuxnet

~1000 centrifuges  
multi-year program

## Direct + indirect

Recovery costs typically exceed ransom by 5–10×. The lasting damage is reputation and regulatory.

*Source:* Colonial: FBI 2021; Norsk Hydro Q1 2019 report; Maersk Q2 2017 interim report; Stuxnet centrifuge count: D. Albright et al., ISIS, 2010.



## New attack surface

Every cloud connector is a new ingress – and a new authentication challenge for legacy field devices.



- **OT Security Engineer** – design, deploy, monitor controls; bridges IT and OT
- **Control Engineer with security focus** – starts from process side
- **ICS Penetration Tester** – offensive; specialised employers
- **Compliance / Audit** – IEC 62443, NIS2 specialist
- **Incident Responder (OT)** – on-call, often consulting-side (Dragos, Mandiant)
- **Researcher** – vendor / academic; vulnerabilities, threat intel

## Demand

Headhunter reports consistently rank OT security in the top three hardest-to-fill cyber roles globally.

|                              |
|------------------------------|
| ICS & SCADA fundamentals     |
| Network architectures        |
| Industrial protocols         |
| Threat landscape             |
| Vulnerabilities & assessment |
| Risk management              |
| Defense in depth             |
| Standards & governance       |
| Incident response            |

### ★ Key Takeaway: Industrial security is different from IT security

- OT differs fundamentally from IT in goals, lifecycle, and tolerance for failure.
- Availability and safety dominate the CIA triad in industrial contexts.
- Adversaries today span from script kiddies to nation-state APTs.
- The next sections build the toolbox: components, networks, protocols, defenses.

- Albright, D., Brannan, P., Walrond, C. *Did Stuxnet take out 1,000 centrifuges at the Natanz enrichment plant?* ISIS, 22 Dec 2010.
- Falliere, N., Murchu, L., Chien, E. *W32.Stuxnet Dossier*, v1.4. Symantec, Feb 2011.
- FBI. *Statement on the Colonial Pipeline attack*, May 2021.
- GAO. *Critical infrastructure protection: Federal authorities for the Colonial Pipeline response*, GAO-22-104746, 2022.
- Norsk Hydro. *Q1 2019 Interim Report* (cyber-attack cost disclosure).
- A.P. Moller – Maersk. *Q2 2017 Interim Report* (NotPetya cost disclosure).
- NIST. *Cybersecurity Framework 2.0*, Feb 2024.
- ENISA. *Threat Landscape*, annual.
- Knapp, E.D., Langill, J.T. *Industrial Network Security*, 2nd ed., Syngress, 2014.
- Bodungen, C.E. et al. *Hacking Exposed: Industrial Control Systems*, McGraw-Hill, 2016.

# End of Section 01

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Introduction to Industrial Security

Questions and discussion